STATS 200A Lecture 13

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Remark 13.1 (Tweedie's Identity)

$$\boxed{D^\star(y) = \mathbb{E}[X \mid Y = y] = y + \sigma^2 s_Y(y), \qquad s_Y(y) := \frac{d}{dy} \log p_Y(y).}$$

Remark 13.2 (Estimate the denoiser from noisy observations)

$$p_Y \mapsto D^\star(Y) = y + \sigma^2 (\log p_Y)'(y). \qquad \hat{D}(y) = y + \sigma^2 (\log \hat{p}_Y)'(y).$$

Remark 13.3 (Read Tweedie in the other direction)

$$\boxed{s_Y(y) = \frac{D^\star(y) - y}{\sigma^2}, \qquad \hat{s}(y) := \frac{D(y) - y}{\sigma^2}}$$

Remark 13.4 (Predict a function)

$$(Kh)(y) := \int h(x)\, K(y, dx).$$

Average h under the conditional law:

$$(Kh)(Y) = \mathbb{E}[h(X) \mid Y]$$

Remark 13.5 (Average conditional laws)

$$(\nu K)(B) := \int K(y, B)\, \nu(dy)$$

Recover the marginal law

$$\nu K = \mu$$

Duality is total expectation:

$$\boxed{\int Kh\, d\nu = \mathbb{E}[\mathbb{E}[h(X) \mid Y]] = \mathbb{E}[h(X)] = \int h\, d(\nu K).}$$